

CodeAlpha IoT Internship - Task 2

Title

Motion Detection System using Arduino and PIR Sensor

Objective

The objective of this project is to detect human motion using a PIR sensor. When motion is detected, the Arduino turns the LED ON. When no motion is detected, the LED turns OFF.

Components

Arduino Uno

PIR Motion Sensor (HC-SR501)

LED

220 Ohm Resistor

Jumper Wires

Tinkercad

Circuit Connections

PIR VCC -> 5V

PIR GND -> GND

PIR OUT -> D2

LED Anode -> D13 through 220 Ohm resistor

LED Cathode -> GND

Arduino Code

```
int pirPin = 2;
int ledPin = 13;

void setup(){
  pinMode(pirPin, INPUT);
  pinMode(ledPin, OUTPUT);
}
void loop(){
  if(digitalRead(pirPin)==HIGH){
    digitalWrite(ledPin,HIGH);
    delay(3000);
  } else {
    digitalWrite(ledPin,LOW);
  }
}
```

Code Explanation

The PIR sensor is connected to digital pin 2. The LED is connected to digital pin 13. The `setup()` function configures the input and output pins. In `loop()`, Arduino continuously checks the PIR sensor. When motion is detected, the LED turns ON for 3 seconds; otherwise it remains OFF.

Working

- The PIR sensor continuously monitors movement.
- When a person moves, the sensor outputs a **HIGH** signal.
- Arduino reads the signal and turns the LED ON.
- The LED stays ON for **3 seconds**.
- If no motion is detected, the LED turns OFF.

Applications

Home security, Automatic lighting, Smart homes, Office security, Motion alarm systems.

Conclusion

This project demonstrates a simple IoT-based motion detection system using Arduino and a PIR sensor.

Diagram

